

Calculus 1 sec 2: Assignment 4, due Thursday April 16, 2026, 11:59pm

Submit the completed assignment to <https://www.gradescope.com>.

Only hand-written solutions are accepted.

- This assignment is worth a total of 15 points, and accounts for about 3% of your final grade. (The bonus is extra.) The standard submission deadline is Thursday April 16, 11:59pm (with an extra no penalty day.) Submissions after the late due date will NOT be accepted unless you have a good excuse.
- Collaboration is allowed, but you must write up your solutions independently and in your own words. Copying from others, publications, or websites (even with minor changes) will be treated as plagiarism.
- Neatness, clarity, and correct notation matter! Show all steps, use clear explanations, and write in complete sentences where needed. Graders will only evaluate what you write, so unclear or incomplete work will affect your grade, even with the correct answer.
- Attempt all questions (and parts), and clearly indicate which question you are answering. Answer each question on a separate sheet, and please **match your answers with the correct questions on Gradescope**.

1 (i) [2 pt] Let f be a twice differentiable function with $f'(x) > 0$ and $f''(x) > 0$ for all x .

a) Can f have a horizontal asymptote at $-\infty$?

b) Can f have a horizontal asymptote at ∞ ?

Explain why or why not. If the answer is "yes" give an example of a function with this behavior.

(ii) [1pt] Give an example of a function f such that $\lim_{x \rightarrow \infty} f(x) = \infty$, and there is a such that $f'(x) > 0$, $f''(x) < 0$ for all $x > a$.

(iii) [2pt] Sketch the graph of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ with the following properties:

$f(x) > 0$ for all x , $f(0) = 3$, f has absolute minimum at $x = -1$ and absolute maximum at $x = 2$, f has no horizontal asymptotes.

2. (i) [3pt] Show that the function $f(x) = e^{-x} \ln(x)$ has a critical number in the interval $[1, 3]$. (Use a theorem rather than a calculator to justify your answer. To do this you must state the theorem and verify its hypotheses.)

(ii) [2pt] Is this number a local max? a local min? neither? Justify your answer.

3. [5 points] Follow the steps A to H listed below to graph the function $f(x) = \sin(2x) + \cos(2x)$ over the interval $0 \leq x \leq 2\pi$.

Bonus. [5 points] Follow the steps A to H listed below to graph the function $f(x) = \frac{e^x}{1+x}$.

- Find the domain of f .
- Find the x - and y -intercepts of f .
- Determine whether or not the graph of f presents any symmetries (i.e. whether or not f is even, odd, or periodic).
- Find the asymptotes of f (check for horizontal, and vertical asymptotes).
- Find the intervals of increase or decrease of f .
- Find the local extreme values of f .
- Determine the concavity of the graph of f and find the points of inflection, if any.
- Sketch the graph of f . Annotate your plot.